

## Exam Cheating Detection Report

**18 Potential Cheating Incident(s) Detected**

|                     |                  |
|---------------------|------------------|
| Video Duration      | 5m 3s            |
| Original Resolution | 1028x576         |
| Frame Rate          | 30.0 fps         |
| Analysis Date       | 2026-02-02 15:05 |

## Executive Summary

### Analysis Overview:

- Total students identified: 19
- Teacher identified: Yes
- Potential cheating pairs detected: 18

### Alert Summary

| Students        | Confidence | Tier      | Primary Behaviors                                  |
|-----------------|------------|-----------|----------------------------------------------------|
| R1_C13 & R1_C9  | 90.0%      | VERY_HIGH | Body leaning toward peer (2100 instances), Hand... |
| R1_C13 & R1_C14 | 90.0%      | VERY_HIGH | Body leaning toward peer (4150 instances), Hand... |
| R1_C8 & R1_C7   | 88.8%      | VERY_HIGH | Body leaning toward peer (3450 instances), Hand... |
| R1_C9 & R1_C14  | 81.0%      | HIGH      | Body leaning toward peer (3550 instances), Hand... |
| R1_C6 & R1_C7   | 80.4%      | HIGH      | Body leaning toward peer (3000 instances), Hand... |
| R1_C14 & R1_C11 | 68.6%      | MEDIUM    | Body leaning toward peer (5550 instances), Hand... |
| R1_C6 & R1_C8   | 67.4%      | MEDIUM    | Body leaning toward peer (3000 instances), Hand... |
| R1_C17 & R1_C19 | 64.2%      | MEDIUM    | Sustained head turn toward peer (300 instances)... |
| R1_C10 & R1_C13 | 63.7%      | MEDIUM    | Body leaning toward peer (1400 instances), Hand... |
| R1_C10 & R1_C14 | 62.2%      | MEDIUM    | Body leaning toward peer (2900 instances), Hand... |
| R1_C5 & R1_C6   | 60.9%      | MEDIUM    | Body leaning toward peer (2700 instances), Hand... |
| R1_C5 & R1_C7   | 60.0%      | MEDIUM    | Body leaning toward peer (3350 instances), Hand... |
| R1_C10 & R1_C9  | 58.0%      | MEDIUM    | Body leaning toward peer (1350 instances), Hand... |
| R1_C8 & R1_C9   | 55.0%      | MEDIUM    | Body leaning toward peer (2400 instances), Hand... |
| R1_C9 & R1_C7   | 55.0%      | MEDIUM    | Body leaning toward peer (2300 instances), Hand... |
| R1_C14 & R1_C12 | 54.2%      | LOW       | Body leaning toward peer (700 instances), Hand ... |
| R1_C17 & R1_C14 | 54.1%      | LOW       | Body leaning toward peer (2000 instances), Hand... |
| R1_C5 & R1_C9   | 50.7%      | LOW       | Body leaning toward peer (2600 instances), Hand... |

## Detailed Alert Analysis

### Alert 1: R1\_C13 & R1\_C9

**Confidence Score:** 90.0% (VERY\_HIGH)

**Suspicious Time Periods:** 8.0s - 21.4s, 3.8s - 14.4s, 5.8s - 18.0s, 10.0s - 25.6s, 12.6s - 28.4s

**Detected Behaviors:**

- Body leaning toward peer (2100 instances)
  - Hand activity near face (2100 instances)
  - Sustained head turn toward peer (1545 instances)
  - Whispering posture detected (1540 instances)
  - Moved closer than normal (975 instances)
  - Mutual gaze (looking at each other) (558 instances)
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### Alert 2: R1\_C13 & R1\_C14

**Confidence Score:** 90.0% (VERY\_HIGH)

**Suspicious Time Periods:** 292.4s - 302.2s, 183.8s - 223.0s, 290.2s - 300.2s, 186.0s - 228.4s, 288.2s - 298.2s

**Detected Behaviors:**

- Body leaning toward peer (4150 instances)
  - Hand activity near face (4150 instances)
  - Sustained head turn toward peer (3420 instances)
  - Whispering posture detected (3340 instances)
  - Mutual gaze (looking at each other) (1301 instances)
  - Moved closer than normal (1195 instances)
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### Alert 3: R1\_C8 & R1\_C7

**Confidence Score:** 88.8% (VERY\_HIGH)

**Suspicious Time Periods:** 126.6s - 151.4s, 136.8s - 153.4s, 124.6s - 147.4s, 142.2s - 159.4s, 223.6s - 267.0s

**Detected Behaviors:**

- Body leaning toward peer (3450 instances)
  - Hand activity near face (3450 instances)
  - Sustained head turn toward peer (3074 instances)
  - Whispering posture detected (3069 instances)
  - Mutual gaze (looking at each other) (1056 instances)
  - Moved closer than normal (891 instances)
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## Alert 4: R1\_C9 & R1\_C14

**Confidence Score:** 81.0% (HIGH)

**Suspicious Time Periods:** 186.6s - 206.0s, 213.0s - 225.4s, 184.6s - 202.0s, 211.0s - 220.8s, 182.6s - 193.0s

**Detected Behaviors:**

- Body leaning toward peer (3550 instances)
  - Hand activity near face (3545 instances)
  - Sustained head turn toward peer (2828 instances)
  - Whispering posture detected (2813 instances)
  - Mutual gaze (looking at each other) (742 instances)
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## Alert 5: R1\_C6 & R1\_C7

**Confidence Score:** 80.4% (HIGH)

**Suspicious Time Periods:** 115.2s - 126.0s, 112.2s - 124.0s, 117.2s - 128.0s, 110.2s - 121.4s, 108.2s - 119.0s

**Detected Behaviors:**

- Body leaning toward peer (3000 instances)
  - Hand activity near face (3000 instances)
  - Moved closer than normal (1984 instances)
  - Mutual gaze (looking at each other) (1701 instances)
  - Sustained head turn toward peer (1526 instances)
  - Whispering posture detected (1495 instances)
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## Alert 6: R1\_C14 & R1\_C11

**Confidence Score:** 68.6% (MEDIUM)

**Suspicious Time Periods:** 5.8s - 26.2s, 7.8s - 28.2s, 9.8s - 30.2s, 3.8s - 24.2s, 237.6s - 247.4s

**Detected Behaviors:**

- Body leaning toward peer (5550 instances)
  - Hand activity near face (4265 instances)
  - Moved closer than normal (3482 instances)
  - Sustained head turn toward peer (3144 instances)
  - Whispering posture detected (2484 instances)
  - Mutual gaze (looking at each other) (1669 instances)
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## Alert 7: R1\_C6 & R1\_C8

**Confidence Score:** 67.4% (MEDIUM)

**Suspicious Time Periods:** 123.0s - 143.4s, 125.0s - 146.8s, 120.0s - 141.4s, 100.0s - 115.0s, 102.0s - 117.8s

**Detected Behaviors:**

- Body leaning toward peer (3000 instances)
  - Hand activity near face (3000 instances)
  - Sustained head turn toward peer (1518 instances)
  - Whispering posture detected (1481 instances)
  - Mutual gaze (looking at each other) (1457 instances)
  - Moved closer than normal (199 instances)
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## Alert 8: R1\_C17 & R1\_C19

**Confidence Score:** 64.2% (MEDIUM)

**Suspicious Time Periods:** 26.8s - 47.6s, 12.8s - 45.2s, 29.4s - 50.6s, 43.4s - 54.8s, 45.4s - 57.0s

**Detected Behaviors:**

- Sustained head turn toward peer (300 instances)
  - Whispering posture detected (300 instances)
  - Body leaning toward peer (300 instances)
  - Hand activity near face (300 instances)
  - Moved closer than normal (222 instances)
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## Alert 9: R1\_C10 & R1\_C13

**Confidence Score:** 63.7% (MEDIUM)

**Suspicious Time Periods:** 137.2s - 148.8s, 135.2s - 145.4s, 34.0s - 65.0s, 126.0s - 143.2s, 27.6s - 43.6s

**Detected Behaviors:**

- Body leaning toward peer (1400 instances)
  - Hand activity near face (1392 instances)
  - Sustained head turn toward peer (1150 instances)
  - Whispering posture detected (1107 instances)
  - Moved closer than normal (922 instances)
  - Mutual gaze (looking at each other) (196 instances)
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## Alert 10: R1\_C10 & R1\_C14

**Confidence Score:** 62.2% (MEDIUM)

**Suspicious Time Periods:** 87.2s - 97.0s, 89.2s - 99.0s, 85.2s - 95.0s, 91.2s - 101.0s, 109.2s - 120.6s

**Detected Behaviors:**

- Body leaning toward peer (2900 instances)
  - Hand activity near face (1958 instances)
  - Sustained head turn toward peer (1289 instances)
  - Whispering posture detected (881 instances)
  - Moved closer than normal (463 instances)
  - Mutual gaze (looking at each other) (414 instances)
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## **Alert 11: R1\_C5 & R1\_C6**

**Confidence Score:** 60.9% (MEDIUM)

**Suspicious Time Periods:** 185.6s - 205.6s, 60.8s - 86.2s, 191.4s - 210.6s, 54.0s - 73.2s, 51.4s - 70.6s

**Detected Behaviors:**

- Body leaning toward peer (2700 instances)
  - Hand activity near face (2700 instances)
  - Sustained head turn toward peer (1396 instances)
  - Whispering posture detected (1381 instances)
  - Mutual gaze (looking at each other) (1316 instances)
  - Moved closer than normal (72 instances)
  - Paper/material sharing gesture (6 instances)
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## **Alert 12: R1\_C5 & R1\_C7**

**Confidence Score:** 60.0% (MEDIUM)

**Suspicious Time Periods:** 127.8s - 142.6s, 129.8s - 144.6s, 131.8s - 147.0s, 125.8s - 139.8s, 9.4s - 22.2s

**Detected Behaviors:**

- Body leaning toward peer (3350 instances)
  - Hand activity near face (3350 instances)
  - Sustained head turn toward peer (1723 instances)
  - Whispering posture detected (1723 instances)
  - Mutual gaze (looking at each other) (554 instances)
  - Moved closer than normal (39 instances)
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## **Alert 13: R1\_C10 & R1\_C9**

**Confidence Score:** 58.0% (MEDIUM)

**Suspicious Time Periods:** 88.4s - 101.8s, 141.2s - 158.6s, 133.4s - 149.4s, 3.8s - 46.2s, 137.4s - 152.6s

**Detected Behaviors:**

- Body leaning toward peer (1350 instances)
  - Hand activity near face (1348 instances)
  - Sustained head turn toward peer (800 instances)
  - Whispering posture detected (787 instances)
  - Mutual gaze (looking at each other) (389 instances)
  - Moved closer than normal (67 instances)
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## **Alert 14: R1\_C8 & R1\_C9**

**Confidence Score:** 55.0% (MEDIUM)

**Suspicious Time Periods:** 3.8s - 15.6s, 5.8s - 21.0s, 71.4s - 89.0s, 47.2s - 87.0s, 190.6s - 219.8s

**Detected Behaviors:**

- Body leaning toward peer (2400 instances)
  - Hand activity near face (2400 instances)
  - Moved closer than normal (2028 instances)
  - Mutual gaze (looking at each other) (1286 instances)
  - Sustained head turn toward peer (250 instances)
  - Whispering posture detected (200 instances)
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## **Alert 15: R1\_C9 & R1\_C7**

**Confidence Score:** 55.0% (MEDIUM)

**Suspicious Time Periods:** 3.8s - 14.0s, 5.8s - 16.0s, 8.2s - 21.2s, 10.2s - 25.2s, 211.0s - 230.6s

**Detected Behaviors:**

- Body leaning toward peer (2300 instances)
  - Hand activity near face (2300 instances)
  - Mutual gaze (looking at each other) (1615 instances)
  - Moved closer than normal (350 instances)
  - Sustained head turn toward peer (15 instances)
  - Whispering posture detected (15 instances)
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## **Alert 16: R1\_C14 & R1\_C12**

**Confidence Score:** 54.2% (LOW)

**Suspicious Time Periods:** 91.8s - 245.8s, 287.2s - 302.2s, 197.6s - 264.8s, 285.2s - 300.2s, 268.6s - 285.0s

**Detected Behaviors:**

- Body leaning toward peer (700 instances)
  - Hand activity near face (700 instances)
  - Mutual gaze (looking at each other) (477 instances)
  - Sustained head turn toward peer (191 instances)
  - Whispering posture detected (158 instances)
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## **Alert 17: R1\_C17 & R1\_C14**

**Confidence Score:** 54.1% (LOW)

**Suspicious Time Periods:** 172.8s - 251.4s, 155.2s - 238.2s, 291.2s - 301.6s, 117.2s - 155.0s, 126.4s - 185.2s

**Detected Behaviors:**

- Body leaning toward peer (2000 instances)
  - Hand activity near face (1995 instances)
  - Moved closer than normal (1490 instances)
  - Mutual gaze (looking at each other) (456 instances)
  - Sustained head turn toward peer (358 instances)
  - Whispering posture detected (260 instances)
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## **Alert 18: R1\_C5 & R1\_C9**

**Confidence Score:** 50.7% (LOW)

**Suspicious Time Periods:** 234.4s - 247.6s, 217.4s - 231.2s, 231.4s - 244.8s, 226.4s - 239.2s, 228.6s - 241.2s

**Detected Behaviors:**

- Body leaning toward peer (2600 instances)
- Hand activity near face (2600 instances)
- Mutual gaze (looking at each other) (1985 instances)
- Moved closer than normal (540 instances)
- Sustained head turn toward peer (21 instances)
- Whispering posture detected (21 instances)

# Methodology

This analysis was conducted using a 7-stage computer vision pipeline:

## 1. Video Preprocessing

Frames extracted at 5 fps with CLAHE contrast enhancement in LAB color space.

## 2. Person Detection & Tracking

YOLOv8 object detection with BoT-SORT multi-object tracking for consistent student identification throughout the video.

## 3. Pose Estimation

YOLOv8-Pose model extracts 17 COCO body keypoints. Head orientation is computed using solvePnP with facial landmarks.

## 4. Spatial Analysis

Automatic seat assignment based on median positions during warm-up period. Neighbor graphs built using distance thresholds. Teachers identified by movement patterns and standing posture.

## 5. Behavioral Feature Extraction

Per-frame features include: head direction, mutual gaze detection, whispering posture, body lean, hand position relative to face and neighbors, paper sharing gestures.

## 6. Temporal Classification

Sliding window analysis (10-second windows, 2-second stride) with weighted scoring. Multi-criteria filter requires: minimum suspicious windows, confidence threshold, mutual behavior evidence, and absence of nearby teacher.

## 7. Alert Generation

Confidence tiers (Very High > 85%, High > 70%, Medium > 55%) with timestamped evidence and behavior summaries.

### Important Notes:

- This system is designed to flag potential incidents for human review.
- False positives may occur due to normal classroom interactions.
- Results should be verified by reviewing the evidence frames and video.
- Teacher proximity during flagged periods may indicate supervised activity.